

Making a Digital Double of Alan Shepard's Space Suit

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Figure 1: Full resolution digital double of Alan Shepard space suit, coming in the form of a curvature-adaptive surface mesh textured with 2D maps storing the pointwise properties of a spatially-varying bidirectional reflectance distribution function.

ABSTRACT

In this talk, we explain how we captured the shape and appearance of Alan Shepard's space suit, a challenging asset for any 3D capture technology. Focusing on high-resolution meso-structure reproduction we introduce a new photogrammetry-based hybrid approach to normal/displacement capture, where each image is enriched with normal and depth neural estimations, which is then fused with the output of a traditional photogrammetry pipeline, capturing rich geometric details, in the highly versatile form of material maps.

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1 INTRODUCTION

On May 5th, 1961, the United States of America launched the very first manned spaceflight, during which astronaut Alan Shepard spent 15 minutes at sub-orbital altitude. During this flight, he was wearing a Mercury space suit, which was transferred from NASA to the Smithsonian in 1977 and is currently on display at the National Air and Space Museum.

In this work, we combine traditional photogrammetry and neural material reconstruction to create a high-quality digital double of this unique artefact, with the objective to create a 3D surface mesh equipped with a physically-based material model exhibiting the highest possible fidelity for both cultural heritage preservation and general public online access. In particular, we describe our full capture pipeline and focus on a new hybrid normal and displacement fusion method that improves meso-structure and fine detail recovery in the digital double.

2 THE SUIT

This is the space suit worn by astronaut Alan Shepard during the first manned spaceflight launched by the United States. Mercury space suits like this one were a modified version of the B.F. Goodrich Company's Mark IV pressure suit, with the most obvious alteration being the aluminized thermal coating which gave the space suit its signature silver appearance. The suit was digitized in 2022 as part of the National Air and Space Museum's "Reboot the Suit" initiative with the goal of making a digital twin of this iconic object freely

available for research, education, and general public access. The digital twin is also a snapshot of the suit's current state which can be used as a baseline for ongoing conservation efforts.

3 CAPTURE

Images were captured by a Canon 5DSR camera with a Canon L Series 35mm lens. The assembly was mounted on a tripod with a wheeled base for easy positioning. A concentric ring with marks every ten degrees was drawn on the floor around the space suit to ensure consistent imaging. The camera was moved up at regular intervals between each ring of 36 images, resulting in a roughly 120-degree arc of coverage vertically around the space suit and full 360-degree coverage around the object. A laser range finder was used to maintain a consistent distance from the object when the camera's height was adjusted. Three sets of images were captured in this manner: two using a light diffusion tent, and one using direct lighting from a ring flash. No attempt was made to match camera position between the capture sets.

Our light diffusion tent was made of four sides and one top panel fitted with 1.25 stop diffusion material, the floor and vertical corners were lined with bounce material. Two image sets were captured in the tent, diffuse ambient (397 images) and diffuse flash (878 images). For the diffuse ambient image set, lighting was provided by the overhead lights of the large warehouse room where the capture took place. This light was diffused off the white walls of the room and then further diffused by the light tent. For the diffuse flash images, illumination was provided by four Broncolor Unilite flash heads, one per side, set back 150 cm from the tent and fitted with large 120 x 180 cm Broncolor softboxes, each fitted with three layers of diffusion material. A self-leveling 3-axis laser line level was used during the whole process for a precise and consistent layout.

The ring flash capture (448 images) used a Broncolor RingFlash mounted to the camera. The ring flash and camera lens were cross-polarized to cancel out direct light reflections (specular reflections) from the suit. To minimize secondary light bounces from external sources, these captures were conducted in a large room with high ceilings and the floor around the suit was covered in light absorbing black velvet. Secondary bounces from the suit itself were unavoidable.

4 RECONSTRUCTION

Based on the captured data, we reconstructed a digital double in the form of an adaptive mesh, equipped with a material coming in the form of a spatially-varying microfacet BRDF based on the GGX normal distribution function and modeled as a collection of 2D texture maps, including normal and displacement components



Figure 2: Left: interior of light diffusion tent. Right: example of capture diffuse flash (in tent), diffuse ambient (in tent), and cross-polarized (no tent).

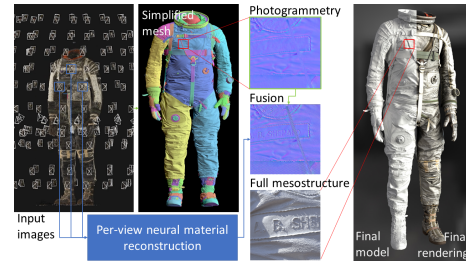


Figure 3: Hybrid reconstruction: per-view normal and displacement reconstruction fused with the base surface.

capturing the meso-structures of the suit. We started by generating a high-resolution (500M triangles) photogrammetry mesh using diffuse lighting photos only, to avoid strong cast shadows and high-lights stemming from flash photos that typically cause geometric artifacts. We then decimated the mesh using an adaptive simplification algorithm based on the quadric error metric, preserving sharp features and high curvature regions. The resulting mesh (400k triangles) was clustered based on its normal vector field, yielding near-planar regions that were parameterized and stored into 11 Udim for efficient editing. With this base surface domain in hand, we then went back to the full captured data set, and registered each image w.r.t. the mesh before projecting their content into our Udim at 8K resolution each. Our approach to the material reconstruction works in two steps: first we estimate a PBR material through a neural material extraction [Martin et al. 2022] for each single view, recovering the fine geometric details (normal/displacement map) typically lost when using photo-consistency; second, we combine these micro details with the photogrammetry normal field using the approach of Barré-Brisebois and Hill [2012], and act similarly for the displacement component. Last, we extract the roughness map by operating a texture-space specular/diffuse separation based on the difference between diffuse and cross-polarized captures, computing the diffuse albedo map from the complete diffuse set and estimating the specular color map from the flash set.

5 RESULTS & DISCUSSION

The resulting adaptive mesh together with its hybrid material model, can be rendered in real-time using GPU path tracing. By transferring the bulk of the meso-structure into highquality normal and displacement maps, we can leverage recent high fidelity rendering techniques, such as *Tessellation-Free Displacement Mapping for Ray Tracing* [Thonat et al. 2021], but also cope with online visualization, as material mipmapping provide easy, graceful degradation free from explicit geometric level-of-detail computations. In the future, we plan to explore a fine tuning strategy for our neural material reconstruction model, to specialize it for domain-specific scenarios.

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